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PI Name:	Turner, Russell T Ph.D.		
Project Title:	Housing Temperature: An Important Variable for Simulated Spaceflight Studies Using Mice		
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Program/Discipline--Element/Subdiscipline:			
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
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COI Name (Institution):	Branscum, Adam Ph.D. (Oregon State University) Iwaniec, Urszula Ph.D. (Oregon State University) Wong, Carmen Ph.D. (Oregon State University)		
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Task Description:	Spaceflight, by altering the differentiation program of hematopoietic and mesenchymal stem cells residing within bone marrow, results in bone loss, increased bone marrow adiposity, anemia, and impaired immune function. These closely associated disturbances may compromise the success of long-term missions. Thus, there exists an urgent need to identify the underlying mechanisms and implement effective countermeasures. Mice are currently the preferred animal model for evaluating adaptive responses to microgravity experienced during spaceflight and simulated spaceflight (e.g., hindlimb unloading). Mice have important advantages over other animal models for spaceflight studies, including small size and ease of genetic manipulation. Although mice and humans share many common characteristics, fundamental species differences in thermoregulation may contraindicate the mouse, as currently applied, as a model for human spaceflight. Humans are homeotherms and when exposed to a cold environment defend their core body temperature, whereas mice are obligatory daily heterotherms and experience cyclic changes in core temperature when subjected to cold stress (i.e., temperature below thermoneutral). Mice are typically housed at or near room temperature (~22°C), which is well below the thermoneutral zone for the species (~32°C). Therefore, mice must expend energy to maintain core body temperature. Cold stress induced by sub-thermoneutral housing increases sympathetic outflow to peripheral tissues, including brown adipose tissue, and has profound effects on metabolism. We have recently shown that cold stress induced by room temperature housing results in rapid cancellous bone loss in mice. Based on this finding, we hypothesize that activation of adaptive thermogenesis in mice housed at room temperature introduces unrecognized and uncontrolled confounding variables into mouse studies. Strategies used by weight-bearing mice to minimize heat loss during room temperature housing (e.g., huddling or postural adjustments) are less effective during spaceflight and simulated spaceflight. This results in increased dependence on adaptive thermogenesis, likely exaggerating the negative physiological effects of skeletal unloading on bone and immune cells. Additionally, the thermogenic mechanisms mediating cold stress-induced changes in metabolism in mice are unlikely to be directly translatable to astronauts and could therefore confound interpretation of experimental results as applicable to humans. This proposal will explore the individual and combined effects of mild cold stress induced by room temperature housing and hindlimb unloading (HLU) on the skeleton in C57BL6 (B6) mice, a strain commonly used in spaceflight and HLU studies. To accomplish our goal, we propose 2 Specific Aims: Specific Aim 1: Determine the contribution of increased adaptive thermogenesis to bone loss during HLU in mice housed at room temperature. We will accomplish this aim by comparing HLU-induced bone loss in mice housed at room temperature (22°C) with mice housed at thermoneutral (32°C). Specific Aim 2: Determine the lowest sub-thermoneutral housing temperature able to prevent adaptive thermogenesis-associated bone loss. We will accomplish this aim by performing temperature response studies (20-32°C) to evaluate induction of adaptive thermogenesis. Once we identify the lowest sub-thermoneutral housing temperature that does not induce adaptive thermogenesis, we will perform a long-term study to verify that minimal premature bone loss occurs and that magnitude of HLU-induced bone loss does not differ from mice housed at thermoneutral. Successful completion of the proposed research will provide guidance to investigators for insuring housing conditions minimize the confounding effects of species-specific differences in thermoregulation on experimental outcomes with the ultimate goal of optimizing the mouse to model human responses to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One of our goals is to establish mouse models that more accurately replicate human physiology. Mice, because of their small size, short lifespan, and ease of genetic manipulation, are a mainstay for cutting edge research in fundamental skeletal biology, and are increasingly being used as preclinical models for skeletal disuse and metabolic bone disease. There is, however, a critical need to identify and address major limitations of mice as models for human physiology. For example, in contrast to humans, mice experience bone- and bone compartment-specific cancellous bone loss while growing. This premature age-related bone loss is a concern because the most common metabolic bone diseases preferentially target the aged or during conditions such as long duration spaceflight believed to accelerate aging. An important difference between mouse and human physiology that may contribute to premature age-related bone loss is the tight coupling of bone metabolism to thermoregulation in mice. If our hypothesis is correct, mild temperature stress induced by room temperature housing is responsible for premature cancellous bone loss in mice and factors that influence thermoregulation in mice may further compromise studies designed to model human responses by independently altering bone metabolism.
	Progress on Specific Aim 1: Determine the contribution of increased adaptive thermogenesis to bone loss during HLU in mice housed at room temperature. We evaluated the impact of housing temperature on skeletal responses to HLU. The findings described below demonstrate remarkable differences in bone and immune response to HLU between growing mice housed at 22°C and those housed at 32°C. Immune response: For this, we evaluated immune cell subtypes in the spleen and carried out ex vivo stimulation of splenocytes with media control, Lipopolysaccharide (LPS), concanavalin A (ConA), and anti- CD3/CD28 to induce cell proliferation and cytokine production. Production of Th1/Th2/Th17 cytokines were measured (IL10, IL17, Tumor Necrosis Factor alpha (TNF alpha), Interferon-gamma (IFNG), IL 6, IL 4, and IL 2). We detected IL 2, IFNG, and IL6. For each of these cytokines, cold stress induced by room temperature housing exaggerated the immune cell response to a T- cell mitogen following HLU. These data will be included in a future publication. Bone mass and microarchitecture: The femur is an important weight bearing bone that is unloaded during HLU. We evaluated the effects of HLU on total femur and cancellous bone volume fraction in the distal femur metaphysis in growing female mice housed at 22°C and mice housed at 32°C. Compared to baseline, HLU resulted in lower total femur bone volume in mice housed at either temperature, but the reduction was ~2 fold greater (22 % versus 10 %) in mice housed at 22°C. The influence of housing temperature on cancellous bone response to HLU was even more profound. HLU resulted in relative osteopenia at the distal femur metaphysis in mice housed at 32°C by preventing bone

Task Progress:	accrual. This finding is in marked contrast to mice housed at 22°C, where the weight bearing controls failed to accrue additional bone and HLU mice lost cancellous bone. The humerus is a weight bearing bone in the mouse. The femur is unloaded during HLU but the humerus should be subjected to near normal loading. Compared to baseline, HLU had minimal effect on total humerus bone volume in female mice housed at 32°C (-3% change), but resulted in bone loss in mice housed at 22°C (-11% change). These findings support the concept that factors in addition to reduced weight bearing contribute to bone loss in HLU mice housed at 22°C. Studies evaluating the impact of room temperature induced cold stress on skeletal response to HLU on growing mice are published. Similar studies in skeletally mature mice housed at room temperature and thermoneutral have been completed and we are now in the process of manuscript preparation. We performed studies to further define the time course for bone loss in mice housed at room temperature. Importantly, transferring mice to thermoneutral prevents additional bone loss but does not restore bone that had been lost during room temperature housing. Consequently, to prevent cold temperature induced bone changes, it is important to begin housing mice at thermoneutral as early as possible. We initiated studies to further define the mechanisms mediating premature bone loss induced by room temperature housing. Our initial focus was on the role of the sympathetic nervous system. This is because sympathetic outflow is increased at sub-thermoneutral temperatures. In the study summarized below, we evaluated the effect of the ß-blocker propranolol on the skeleton in female B6 mice housed at room temperature (22°C) and thermoneutral (32°C). The work is complete, and a manuscript published. Progress on Specific Aim 2: Determine the lowest sub-thermoneutral housing temperature able to prevent adaptive thermogenesis-associated bone loss. We proposed to accomplish this aim by performing temperature response studies (20°C to 32°C) to measure adaptive thermogenesis as a function of housing temperature (Sub-Aim 1). Once we identified the lowest sub-thermoneutral housing temperature that does not induce adaptive thermogenesis and premature bone loss, we proposed to perform a long-duration study to verify that the magnitude of HLU-induced bone loss in mice housed at this temperature does not differ from mice housed at thermoneutral (Sub-Aim 2). We previously reported that the premature (prior to cessation of growth) cancellous bone loss in female B6 mice housed conventionally (room temperature) was prevented when the mice were housed at thermoneutral (32°C). We performed additional studies to determine whether these findings apply to males. Specifically, we compared growing mice housed at room temperature (22°C) to mice housed at 32°C. The work is complete, and a manuscript published. We performed a serious of studies to establish the lowest sub-thermoneutral housing temperature able to prevent adaptive thermogenesis-associated bone loss., including a long duration study where male and female growing mice were housed at either 22°C or 26°C until adulthood. We chose 26°C because this is the highest temperature recommended in animal care and use guidelines. We have completed collection of data related to energy expenditure, oxygen consumption, carbon dioxide emission, respiratory exchange ratio, food consumption, and distance traveled, and are preparing a publication. Additionally, we analyzed bone mass, density, and microarchitecture. The data clearly indicate that housing mice at 26°C modestly reduced but did not prevent adaptive thermogenesis or premature bone loss. Compared to either baseline or mice housed at 32°C, housing male mice at 26°C marginally reduced premature cancellous bone loss. Similarly, bone loss was reduced but not prevented in female mice housed at 26°C. However, it is important to note that a 4°C difference in housing temperature was sufficient to significantly alter cancellous bone volume fraction in growing male and female B6 mice. In addition, we have performed analysis of brown and white adipose tissue of mice flown and sacrificed aboard the International Space Station (ISS). The results support our hypothesis that non-shivering thermogenesis is increased aboard the ISS. A manuscript describing this research is published. COVID-19 delayed our progress on the funded proposal. However, it provided us an opportunity to evaluate archival tissue and unpublished data to explore generalizability of published results and test a novel hypothesis. In the first case, we evaluated the effects of spaceflight on the femoral head using tissue from two (2) space shuttle experiments. This manuscript is now published. In the second case, we used unpublished data from numerous spaceflight and ground-based studies to test the gravitostat hypothesis for weight control. This manuscript is now published. In the third case, Nrf2 signaling has been implicated in mediating some of the detrimental effects of microgravity on the musculoskeletal system. We therefore participated in a collaborative study evaluating bone from mice in which Nrf2 was deleted. This manuscript is now published. Summary (Interpretation) of Results Specific Aim 1: We have strong evidence that adaptation to cold stress is responsible for ~50% of bone loss in the femur of growing HLU mice. When we consider the cancellous compartment, the impact of housing temperature is even more striking; HLU prevented bone accrual at 32°C and resulted in bone loss at 22°C. In contrast, HLU prevented bone accrual in mice housed at 32°C but did not induce cancellous bone loss. Even more striking was our finding that HLU-induced bone loss in humerus, a bone subjected to near normal weight bearing in the HLU model. Taken together, these findings provide evidence that adaptation to cold stress induced by room temperature housing is an important modifier of the skeletal response to HLU in mice. In addition, analyses of splenocytes suggest that increased cytokine secretion may contribute to excess bone loss induced by HLU in growing room temperature housed mice. We have extended these studies to skeletally mature mice, which show similarly dramatic housing temperature associated differences in response to HLU. We are in the process of preparing these data for publication. Specific Aim 2: Room temperature housing-induced premature cancellous bone loss in growing mice is not sex specific; although male mice have higher peak bone mass, the magnitude of bone loss in room temperature housed mice did not differ between male and female mice. Housing mice at 26°C reduced, but did not prevent, premature cancellous bone loss in either male or female growing mice. However, the observed differences between mice housed at 22°C and 26°C indicate that even small differences in housing temperature influence bone mass in growing mice.
Bibliography Type:	Description: (Last Updated: 01/12/2026)

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